

## **DATABASE SYSTEMS-BIT1223/BTL1223/BCL1223**

### **TUTORIAL 2**

#### **Question 1**

- a) Define the relational database model. (2 marks)
- b) Explain TWO advantages of the relational database model compared to hierarchical and network database models. (4 marks)
- c) Differentiate between logical data independence and physical data independence. (4 marks)

#### **Question 2**

A university maintains the following table:

**STUDENT(StudentID, StudentName, Programme, CGPA)**

- a) Explain the meaning of the terms **relation**, **tuple**, and **attribute** using the STUDENT table above. (6 marks)
- b) Why is a table considered a logical representation of a relation? (4 marks)

#### **Question 3**

Consider the following relation:

**EMPLOYEE(EmpID, EmpName, Department, Salary)**

Given that:

**EmpID → EmpName, Department, Salary**

- a) Define functional dependency. (2 marks)
- b) Explain why EmpID functionally determines the other attributes. (4 marks)
- c) Determine whether EmpName can functionally determine EmpID. Justify your answer. (4 marks)

#### Question 4

Consider the relation:

**COURSE\_REGISTRATION(StudentID, CourseID, Semester, Grade)**

a) Define the following terms:

- i. Superkey
- ii. Candidate Key
- iii. Primary Key
- iv. Composite Key (8 marks)

b) Identify the most appropriate primary key for the relation and justify your answer. (4 marks)

#### Question 5

Given the following tables:

**DEPARTMENT(DeptID, DeptName)**

**EMPLOYEE(EmpID, EmpName, DeptID)**

a) Identify the primary key and foreign key in the tables above. (4 marks)

b) Explain the concept of referential integrity. (3 marks)

c) Describe what would happen if an employee record contains a DeptID that does not exist in the DEPARTMENT table. (3 marks)

#### Question 6

A company stores the following information:

| CustomerID | CustomerName | OrderID | OrderDate   |
|------------|--------------|---------|-------------|
| C001       | Ali          | O101    | 10-Jan-2026 |
| C002       | Siti         | O102    | 12-Jan-2026 |

a) Convert the information into relational schemas. (4 marks)

b) Underline the primary keys and indicate foreign keys. (4 marks)

c) Explain the relationship between the entities identified. (2 marks)

### Question 7

- a) Differentiate between the following relationship types:
- One-to-One (1:1)
  - One-to-Many (1:M)
  - Many-to-Many (M:N) (6 marks)
- b) Provide ONE real-world example for each relationship type. (6 marks)
- c) Explain why M:N relationships cannot be directly implemented in a relational database. (3 marks)

### Question 8

- a) Define a data dictionary. (3 marks)
- b) Explain the role of metadata in a relational database. (3 marks)
- c) Discuss THREE benefits of maintaining a system catalog. (6 marks)

### Question 9

Given the relation:

**STUDENT(StudentID, StudentName, Programme)**

- a) Explain the difference between the SELECT and PROJECT operations. (4 marks)
- b) Write the relational algebra expression to:
- Retrieve all students from the "Computer Science" programme.
  - Display only StudentID and StudentName. (4 marks)
- c) Differentiate between UNION and INTERSECT with suitable examples. (4 marks)

### Question 10

- a) Define data redundancy. (2 marks)
- b) Explain TWO data anomalies that may result from excessive redundancy. (4 marks)
- c) Describe the purpose of indexing in a relational database. (2 marks)
- d) Explain how indexes improve query performance. (4 marks)